

MATHEMATICS 2270

Quiz 7, Fall semester 2012

KEY

Name:.....

1. (5 pts) Let $A = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$. Is A positive definite? What kind of curve corresponds to the equation $2x^2 + 6xy + 2y^2 = 1$?

$$|A| = 2 - 9 = -5. \quad \text{Not pos def}$$

$$\begin{vmatrix} 2-\lambda & 3 \\ 3 & 2-\lambda \end{vmatrix} = \lambda^2 - 4\lambda - 5 = (\lambda - 5)(\lambda + 1) \quad \lambda = 5, -1 \text{ are the eigenvalues.}$$

It is a hyperbola.

2. (5 pts) For what values of c is $A = \begin{pmatrix} 1 & c & 0 \\ c & 1 & c \\ 0 & c & 1 \end{pmatrix}$ positive definite?

$$\begin{vmatrix} 1 & c & 0 \\ c & 1 & c \\ 0 & c & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & c \\ c & 1 \end{vmatrix} = 1 - c^2 > 0 \quad \text{iff} \quad -1 < c < 1$$

$$\begin{vmatrix} 1 & c & 0 \\ c & 1 & c \\ 0 & c & 1 \end{vmatrix} = 1 - c^2 - c^2 = 1 - 2c^2 > 0 \quad \text{iff} \quad -\frac{1}{\sqrt{2}} < c < \frac{1}{\sqrt{2}}$$